Constraints

Creating A Database For An Online Bookstore

```
-- Create Database - creating a database for an online bookstore
CREATE DATABASE Bookstore;
USE Bookstore;
```

Create Table Books and Orders:

```
CREATE TABLE Books(BookID INT PRIMARY KEY AUTO_INCREMENT,Title VARCHAR(255) NOT NULL,
Author VARCHAR(255) NOT NULL,Genre VARCHAR(100),Price DECIMAL(10,2) NOT NULL,
Stock INT NOT NULL,PublishedDate DATE);

CREATE TABLE Orders(OrderID INT PRIMARY KEY AUTO_INCREMENT,BookID INT NOT NULL,
CustomerName VARCHAR(255) NOT NULL,OrderDate DATE NOT NULL,Quantity INT NOT NULL,
TotalAmount DECIMAL(10,2) NOT NULL, FOREIGN KEY (BookID) REFERENCES Books(BookID));

SHOW TABLES;
DESCRIBE Books;
DESCRIBE Orders;
```

Tables_in_bookstore
books
orders

Field	Type	Null	Key	Default	Extra
BookID	int	NO	PRI	NULL	auto_increment
Title	varchar(255)	NO		NULL	
Author	varchar(255)	NO		NULL	
Genre	varchar(100)	YES		NULL	
Price	decimal(10,2)	NO		NULL	
Stock	int	NO		NULL	
PublishedDate	date	YES		NULL	

Field	Type	Null	Key	Default	Extra
OrderID	int	NO	PRI	NULL	auto_increment
BookID	int	NO	MUL	NULL	
CustomerName	varchar(255)	NO		NULL	
OrderDate	date	NO		NULL	
Quantity	int	NO		NULL	
TotalAmount	decimal(10,2)	NO		NULL	

2: UNIQUE Constraint

```sql
ALTER TABLE Books ADD COLUMN ISBN VARCHAR(21) UNIQUE;
DESCRIBE Books;
```

Field	Type	Null	Key	Default	Extra
BookID	int	NO	PRI	NULL	auto_increment
Title	varchar(255)	NO		NULL	
Author	varchar(255)	NO		NULL	
Genre	varchar(100)	YES		NULL	
Price	decimal(10,2)	NO		NULL	
Stock	int	NO		NULL	
PublishedDate	date	YES		NULL	
ISBN	varchar(21)	YES	UNI	NULL	

View tables data:

```sql
-- View data in table
SELECT * FROM BOOKS;
SELECT * FROM ORDERS;
```

BookID	Title	Author	Price	ISBN
1	Data Analytics Essentials	Jane Doe	499.00	9781234567890
2	SQL Mastery	John Smith	599.00	9789876543210
3	Python for Data Science	Alice Brown	699.00	9781111111111
4	Machine Learning Basics	David Clark	799.00	9782222222222
5	Statistics for Analysts	Emily White	549.00	9783333333333
6	Data Visualization with Power BI	Michael Green	599.00	9784444444444
7	Advanced SQL Queries	Sophia Lee	649.00	9785555555555
8	Excel Power Tools	Robert King	399.00	9786666666666
9	Big Data Fundamentals	Laura Scott	899.00	9787777777777
10	Business Intelligence Concepts	Daniel Adams	699.00	9788888888888
11	Data Cleaning Techniques	Olivia Brown	459.00	9789999999999
12	ETL and Data Pipelines	Chris Evans	729.00	9781212121212
13	Cloud Data Warehousing	Rachel Carter	849.00	9781313131313

OrderID	BookID	CustomerName	OrderDate	Quantity
101	1	Narmada	2026-03-18	2
102	2	Rahul	2026-03-18	1
103	3	Priya	2026-03-19	3
104	4	Arun	2026-03-20	1
105	5	Meena	2026-03-20	2
106	6	Karthik	2026-03-21	1
107	7	Divya	2026-03-21	3
108	8	Suresh	2026-03-22	2
109	9	Lakshmi	2026-03-22	1
110	10	Vikram	2026-03-23	4
111	11	Anita	2026-03-23	2
112	12	Ravi	2026-03-24	1
113	13	Geetha	2026-03-24	3

3: DELETE vs TRUNCATE:

```
DELETE FROM Orders where CustomerName = 'Anita';
SELECT * FROM ORDERS;
```

OrderID	BookID	CustomerName	OrderDate	Quantity
101	1	Narmada	2026-03-18	2
102	2	Rahul	2026-03-18	1
103	3	Priya	2026-03-19	3
104	4	Arun	2026-03-20	1
105	5	Meena	2026-03-20	2
106	6	Karthik	2026-03-21	1
107	7	Divya	2026-03-21	3
108	8	Suresh	2026-03-22	2
109	9	Lakshmi	2026-03-22	1
110	10	Vikram	2026-03-23	4
112	12	Ravi	2026-03-24	1
113	13	Geetha	2026-03-24	3

```
TRUNCATE TABLE Orders;
SELECT * FROM ORDERS;
```

OrderID	BookID	CustomerName	OrderDate	Quantity
NULL	NULL	NULL	NULL	NULL